QUICK CARD · TSM2370 MATERIALTEKNIKK

Matsuzawa MXT50 – Vickers HV1

Full guide with photos: **scan the QR code** or go to **ravo10.github.io/USN-kongsberg-lab-mech**. § = section in the guide.

SAFETY

- Diamond and objective must **never** touch the sample
- Do not turn the turret while **LOADING** is lit
- **START** only when the diamond has clicked in
- Raise the stage **very slowly** towards the diamond

A. Start-up §3

1. **ON/OFF** on the right side → ON.
2. Flashlight on, push it fully into the light tube (left side).
3. **Shutter pin** on the right side of the head pushed **IN**.
4. **HV** lamp lit green (if not, press HV/HK).
5. Load dial: **1 Kgf** at the pin.
6. **DWELL TIME** → 15 s, hold **SET 2SEC** for 2 s.

B. Zero the lines §4A

7. Close the lines with the right knob until only a **thin line of light** is left.
8. Hold **SET 2SEC** for 2 s → **d1 = 0**.

D. Check on the HMV700 block §4E

Make **3 indents** on the block (steps 9–16). Expected diagonal ≈ **51 µm**.

✓ Mean **665–735 HV** → the machine is OK.

× Outside → zero the lines, check load and dwell time, measure again.

× Still outside → **tell the lab supervisor**.

Calculation §6

$$HV = 1.8544 \times F / d^2$$

d = (d1 + d2) / 2 in mm, F in kgf. Example: d1 = 51.5 and d2 = 50.7 µm → d = 0.0511 mm → 710 HV1.

Report as e.g. **450 HV1** (mean of 3, with standard deviation).

C. One indent §4B–D, §5

9. 1–2 **support blocks** in the V-block holder, sample on top, polished face up.
10. **Orange objective** (click): find and focus. Raise the stage **slowly** until the surface is sharp.
11. Clean spot with X/Y: ≥ 0.5 mm from edges and old indents.
12. **Diamond** (click). Raise the stage **very slowly** until the tip **almost touches** (look from the side) → **START** → wait for LOADING to go off.
13. **Lower the stage** first (objectives reach lower than the diamond), then **green objective** and focus. Orange reads only ~1/4.
14. **d1:** lines on the left and right corners → **READ**.
15. Turn the eyepiece **90°**. **d2:** top and bottom corners → **READ** → HV under **DATA**.
16. Note d1, d2, HV. Next indent: **one turn** of the X micrometer (0.5 mm).

Troubleshooting §6

Dark image	Flashlight off/flat, or shutter pin out
Blurred corners	Scratched surface – re-polish (≤ 1 µm)
d1 ≠ d2 (> 5 %)	Sample not level
Indent not visible	Turret not clicked – search with X/Y
d ≈ 13 µm, HV too high	Measured with orange – use green

E. Shutdown §6

☐ Orange objective, lower the stage

☐ Take out the sample, block back in its box

☐ Flashlight off and **on charge**

☐ ON/OFF → OFF, dust cover on